



JURONG JUNIOR COLLEGE

JC2 Preliminary Examination 2018

CANDIDATE
NAME

Tr's Copy

CLASS

INDEX
NUMBER

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

10 September 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

1

2

3

Section B

Total

This document consists of **23** printed pages.

Section A

Answer **all** the questions in this section.

- Over thousands and millions of years, there have been natural cycles in the Earth's climate. There have been ice ages (e.g. the Quaternary Ice Age) and warmer interglacial periods. Climatic changes can be investigated using evidence left in tree rings, layers of ice in glaciers, ocean sediments and layers of sedimentary rocks. For example, bubbles of air in glacial ice trap tiny samples of Earth's atmosphere, giving scientists a history of greenhouse gases that stretches back more than 800,000 years.

These changes in climate affect plant distribution and physiology. Fig. 1.1 is a diagram showing the topographical profile of two mountains in the tropics during a warm phase and a cool phase in the Earth's climate. The shape of the lines corresponds to a vertical section through the mountains to show their height and shape. The distribution of rain forest vegetation is also shown.

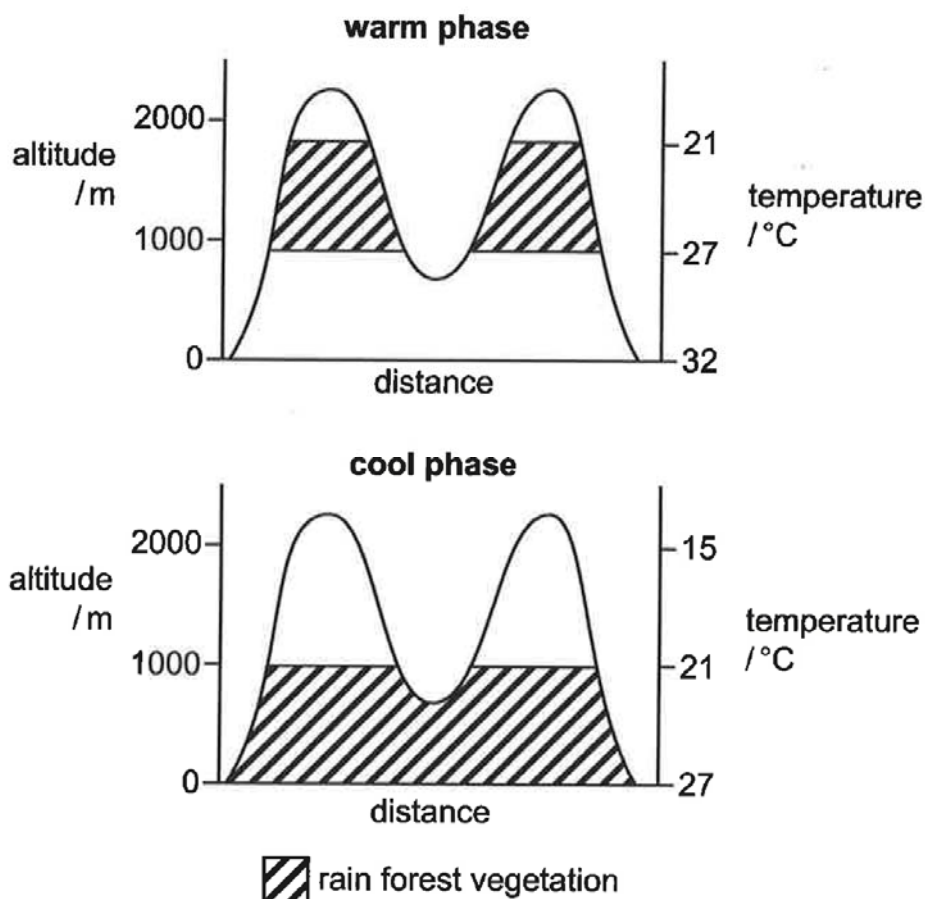


Fig. 1.1

- (a) (i) Describe and explain the effect of climate change on the distribution of rain forest vegetation in the tropics, as shown in Fig. 1.1. [4]

Describe

1. Warm phase 900—1800 m, cold phase 0—1000 m ;
2. Distribution continuous in cool phase, discontinuous/patchy/forming islands in warm phase ;

Explain

3. Rain forest vegetation will only grow in temperature range of 21—27 °C ;
4. Lower temperatures at higher altitude/ORA ;
5. As temperature increases, the distribution of rain forest vegetation shifts towards higher altitudes/ORA ;

- (ii) Over millions of years there are repetitive cycles of climate change resulting in a pattern of alternating warm and cool phases in the Earth's climate.

Suggest how repeated changes in climate between the two phases shown in Fig. 1.1 may lead to evolution of new species and greater species diversity. [4]

Warm phase (max 2)

1. Geographical isolation leading to allopatric speciation ;
2. Reduced/no gene flow/interbreeding ;
3. Different conditions/different selection pressures ;

Cool phase

4. Ref. to spread of new species (to places that they were not prev found in) ;
5. previously isolated populations come back into contact but can no longer interbreed ;
6. ref. to effect of repeated rounds of speciation/fragmentation ;

- (iii) As the atmosphere is thin at the top of the mountains, the plants growing at this region absorb high amounts of light energy from the sun since the sun rays are not scattered or reflected by the air molecules.

Explain how photosynthesis may be affected. [3]

1. At higher light intensities/with more light, there will be higher rate of photosynthesis/light dependent reaction/photophosphorylation/light independent reaction/Calvin cycle ;
2. more electrons of special chlorophyll a in the reaction centre of the photosystems are excited and boosted to a higher energy level/more photoactivation ;
3. increasing movement of electrons down both ETCs ;
4. There is more photolysis of water ;
5. resulting in more ATP and NADPH synthesis ;
6. more triose phosphate and glucose formed ;

any 4 discussion points from MP 2-6 for max 2m

- (b) Changes in the Earth's climate are also expected to have an impact on insects such as the *Aedes aegypti* mosquitoes which are commonly found in the tropics and sub-tropics. As the mosquitoes are the vectors carrying dengue virus and causes dengue in humans, their life-cycles have been studied in detail to manage their population size.

Describe the life-cycle of *Aedes aegypti*. [4]

1. Female *Aedes aegypti* lay their eggs above the water line in areas likely to temporarily flood such as tree holes and containers (e.g. buckets, discarded bottles and flower pots) ;
2. Mosquito larvae hatch from the submerged eggs 2 days after the containers are filled with water and feed on microorganisms found in the water ;
3. Larvae go through developmental stages in which they moult four times over 5 days / moult to become fourth instar ;
4. When a larva is a fully grown fourth instar, it undergoes metamorphosis into a new form called a pupa ;
5. After two days, the fully developed adult mosquito forms and breaks through/emerger from the pupal case ;

1m for 4 correct stages (egg, larva, pupa and adult)

3m for elaborations

1 number of days

- (c) Another mosquito-borne disease of concern worldwide is malaria. Malaria was common in Italy, a European country situated in the Northern Hemisphere. Widespread land drainage together with the use of the insecticide DDT and the drug chloroquine eradicated both the mosquito vectors and the malaria parasites, *Plasmodium*, in the 1950s. Due to the success of these measures, they were later discontinued.

Articles in the scientific literature more recently show that malarial mosquitoes are returning to Italy and increasing their numbers and their northerly range, with some cases of malaria being reported. In general, winters are milder and summers hotter in the south of the country, with temperatures decreasing in the north, especially in winter.

Discuss whether the return of malaria to Italy can be attributed to climate change. [4]

YES

1. Range extending further north may relate to warmer temperatures ;
2. Change/increase, in rainfall may result in more flooded areas for mosquitoes to breed ;
3. Rate of malaria parasite replication within host/vector increases and extrinsic incubation period shortens due to warmer temperatures ;

NO

Any 3 from

4. Mosquitoes, migrate/are introduced, from surrounding countries ;
5. (thrive because they) historically lived in Italy/Italy provides suitable, habitat/conditions, for them ;
6. (thrive because) use of, DDT/some insecticides, was discontinued/banned (after 1950s/eradication) ;
7. Survival and reproduction of malaria parasite in, host/vector, not/hardly, affected by ambient temperature ;

Max 3 if no stand stated (need at least 1) /not balanced

- (d) In many parts of the world, the malaria parasites, *Plasmodium*, have become resistant to chloroquine. In these areas, one of the most effective anti-malarial drugs currently in use is artemisinin. Artemisinin works by binding to an enzyme in *Plasmodium* called PfATP6, acting as an inhibitor.

A substance called curcumin, which has long been used as a spice and yellow food colouring in India and other countries, is also known to act against chloroquine-resistant *Plasmodium*. A group of researchers predicted that curcumin acts by binding to the same enzyme as artemisinin.

In order to test this hypothesis, and to try to find other similar substances that might work better than curcumin, the researchers used theoretical modelling to:

- look at the chemical structures of various molecules with a similar structure to curcumin (curcumin analogues)
- generate a three-dimensional model of the structure of the enzyme PfATP6
- investigate whether each curcumin analogue could bind to PfATP6.

The researchers predicted that several of the curcumin analogues would bind more strongly than curcumin to PfATP6.

- (i) Suggest advantages of using theoretical models in this research, rather than testing possible drugs in the laboratory. [2]

1. **cheaper / more economical / more cost-effective / less wastage leading to cost savings ;**
2. **faster / can try many different drugs in a short time / time-saving / greater ease / theoretical models can reproduce the structure of the analogues that are rare or hard to acquire or produce ; [R: less effort]**
3. **can try out changes to model/drug structure to see if more effective ;**
4. **no need for laboratories/equipment ; IGNORE: uses less labour**
R: resources [Must specifically state what type of resources]
5. **(initially) no need for tests on animals/humans / fewer ethical issues ;**

REJECT:

- **ORA [i.e. why lab methods more disadvantageous]**
- ***“can be quantified and statistical analysis can be done”***
- ***“prevent the development of drug-resistant pathogens”***
- ***“does not involve use of actual malaria parasite and thus reduce risk of disease outbreak from the laboratory”***

(ii) Suggest why theoretical modelling cannot completely replace laboratory trials in the search for new drugs. [2]

1. functionality/to test that drug, actually works/is effective / to test whether drugs are interfered by other molecules *in vivo* or in the parasite / if drug indeed inhibits *Plasmodium* ;
2. A cannot assume predictions are correct ; [I: efficiency, R: based on theory]
3. safety ; A ref. to clinical trials/side effects
4. dosage ; A theoretical modelling will not give information on doses

REJECT:

- Theoretical models only takes into account the spatial fit of the inhibitor in the binding site of PfATP6 and not the chemical fit. [Reason: theory should consider all aspects of fit already]
- Owing to selection for drug resistance, the structure of the allosteric site of PfATP6 had mutated with time and the analogues are no longer complementary to the allosteric site. Thus, theoretical models are unable to keep current or keep up with these new mutations. [Reason: neither can laboratory trials keep up! In fact it is harder]

(e) In places such as Africa where malaria is endemic, sickle cell anaemia — an inherited blood disorder — is prevalent in the populations. Sickle cell anaemia is caused by a single base pair substitution.

(i) Describe the effects of the mutation of the gene involved in sickle cell anaemia. [4]

1. In sickle cell anaemia, a single base pair substitution of the gene coding for β chain causes adenine to replace thymine / DNA triplet to be changed from CTC to CAC in the gene/DNA ;
2. In the mRNA formed, GUG is coded for instead of GAG / resulting in hydrophobic valine replacing hydrophilic glutamic acid (at the 6th amino acid position of β chain) ;
3. This results in a change in the 3D conformation of haemoglobin to produce haemoglobin S (HbS) instead of HbA ;
4. This decreases the solubility of deoxygenated HbS and at low oxygen concentration, hydrophobic areas of different HbS would stick together / HbS molecules will polymerise and precipitate out of solution to form rigid fibres;
5. red blood cells becomes sickle shape and may block capillaries, impeding blood flow / deprives multiple organs of oxygen resulting in organ damage / have shorter lifespan and haemolyse readily, resulting in anaemia ;

- (ii) Only those individuals that inherit two copies of the recessive allele which causes sickle cell anaemia develop the disease. If left untreated, these individuals have a shorter than normal life expectancy. As such, it would be expected that this mutation would be rare in human populations. However, observations made during the mid-20th century revealed that there are sometimes 10-40% of the population carrying this recessive allele.

Explain how there can be a high proportion of the population carrying the recessive allele even when it is disadvantageous. [3]

1. Due to heterozygote advantage ;
2. Individuals who are heterozygous for the recessive allele which causes sickle cell anaemia/sickle cell allele have greater fitness / at selective advantage as compared to individuals who are homozygous ;
[Must demonstrate *relative* advantage over homozygotes (including both $Hb^S Hb^S$ and $Hb^A Hb^A$)]
3. As heterozygotes are protected against a type of severe malaria [R: diseases], hence maintaining genetic variation in the population/ both alleles in the population/ a high frequency of the recessive allele ;

[Total: 30]

- 2 Fig. 2.1 shows stem cells in bone marrow dividing by mitosis to form blood cells. Each time a stem cell divides, it forms a replacement stem cell and a cell that develops into a blood cell.

Stem cells in the bone marrow give rise to reticulocytes, phagocytes, B-cells and T-cells.

Reticulocytes will differentiate into red blood cells.

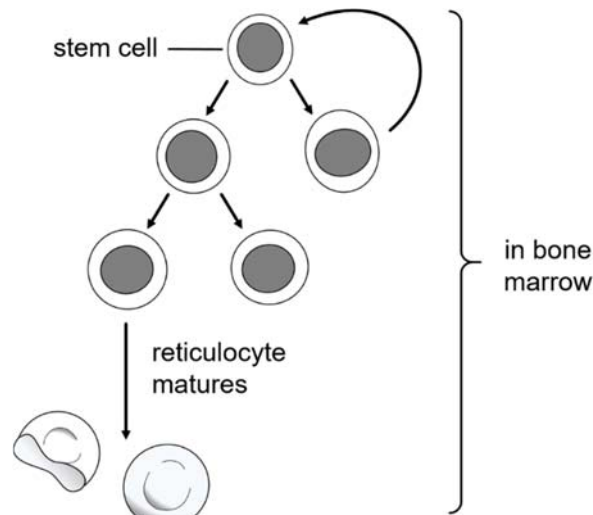


Fig. 2.1

- (a) Describe how a functional red blood cell develops from a stem cell. [3]

1. **transcription (of specific genes) ; A: reference to differential gene expression**
2. **protein / polypeptide, synthesis ; A: translation**
3. **production of haemoglobin ;**
4. **further detail of haemoglobin formation ; e.g. assembly of quaternary structure**
5. **loss of, mitochondria / nucleus ;**
6. **adopts biconcave disc shape ;**
7. **increase in membrane elasticity of rbc to move through capillaries ;**

- (b) Stem cells share some features of cancer cells, such as having long life span and ability to replicate for extended periods of time.

In some cases, the use of stem cells in medical treatment of genetic diseases appears to increase the risk of cancer.

Suggest why there might be a connection between the medical use of stem cells in treatment and increased cancer risk. [2]

1. **Stem cells and cancer cells share similarity of having active telomerase which a) prevents the progressive shortening of telomeres with each round of DNA replication OR b) have limitless replicative potential/divide indefinitely ;**

any 1:

2. **When exposed to carcinogens, spontaneous mutations occurs within stem cells resulting in uncontrolled cell division ;**
3. **Ref to one featur of cancer hallmark accumulated which heightens the risk of of cancer ;**

Stem cells can also differentiate to form T cells. T cells are arguably the most important cells in adaptive immunity, as they are required for almost all adaptive immune responses.

A person infected with the HIV virus was monitored for several years for the level of T cells and HIV particles.

The results are summarised in Fig. 2.2.

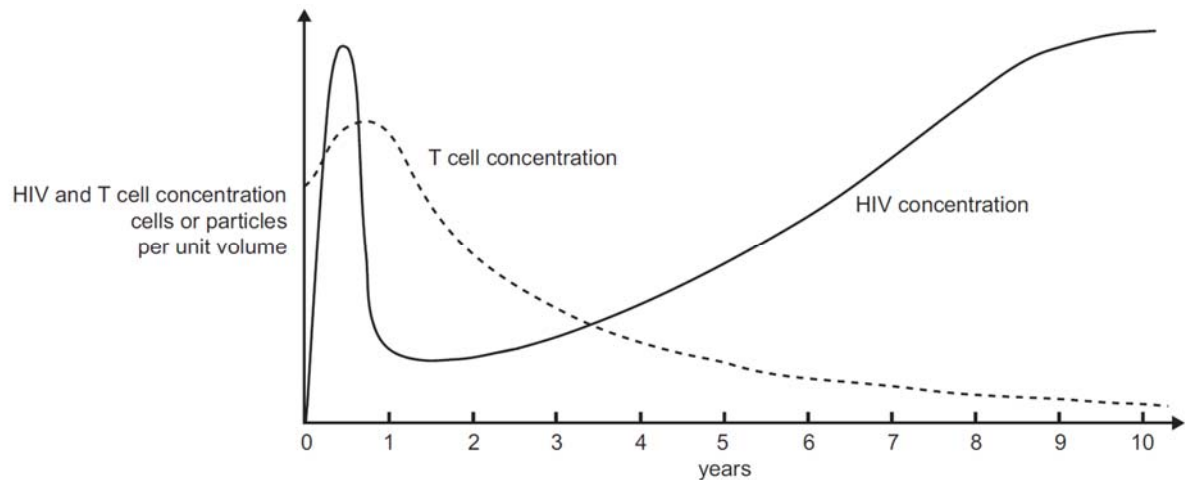


Fig. 2.2

(c) Explain the trend observed from 0.5 to 10 years. [4]

1. **Sharp decrease** in HIV concentration 0.5 to 1 year post-infection ;
2. Concentration of T cells increased to remove HIV at the initial stage of infection
OR
3. No active replication of HIV upon first half year post-infection OR
4. HIV infects T_H cells with some provirus integrating into the chromosome of T_H cells and remain dormant/ latent stage ;
5. HIV concentration **increases gradually** while T cells concentration **decreases gradually** from 1 to 10 years ;
6. Untreated HIV infection led to the replication of new virus that triggers the destruction of T_H cells ;

Enzymes essential in the life cycle of human immunodeficiency virus (HIV) have been at the focus of research. HIV-1 protease is one such enzyme.

HIV-1 protease is formed from two identical polypeptides, each of which is made up of 99 amino acids. The polypeptides both have α helices and β strands. The active site of the enzyme is formed by three amino acids from each polypeptide. Fig. 2.3 shows HIV-1 protease.

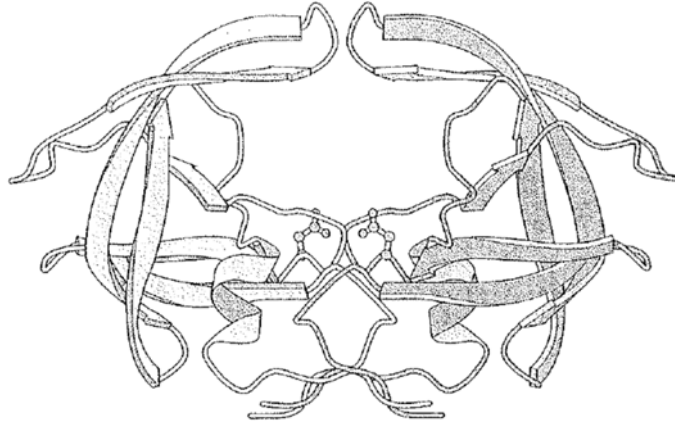


Fig. 2.3

Over the years, knowledge of the structure led to designs of HIV protease-inhibiting drugs with improved antiviral properties. Despite the qualified success of these inhibitors, the high mutation rate associated with RNA viruses continues to hamper the long-term clinical efficacy of HIV protease inhibitors.

(d) Suggest why this means that researchers may need to continue to develop new drugs in order to prevent the action of HIV-1 protease. [2]

1. mutations to the viral genome could alter the genes coding for HIV-1 protease ;
OR
2. change in 3-D conformation of HIV-1 protease's active site ;

any 1:

3. drugs that are previously complementary to the active site may not be complementary to the new active site that resulted ;
4. ref to new strains of HIV that are resistant to drugs ;

[Total: 11]

- 3 G protein-coupled receptors (GPCRs) constitute a large protein family of receptors that detect molecules outside the cell and activate internal signal transduction pathways and, ultimately, cellular responses in mammals.

(a) Describe two ways in which GPCRs are similar in structure and explain why these features are important for their function. [4]

1. (S) has (extracellular) binding site for the (specific) signal molecule/ligand ;
AND
2. (F) recognised and bound by (hydrophilic,) polar ligands (which are unable to pass freely across the membrane) ;
3. (S) the extracellular parts of G-protein linked receptors may be glycosylated ;
AND
4. (F) serve as the binding site for (specific/ complementary) signal molecule/ ligand ;
5. (S) has (intracellular) binding site for G-protein ;
AND
6. (F) the G-protein linked receptor undergoes a conformational change upon the binding of a signal molecule/ligand that causes it to bind and activate a specific G-protein (which activates other signal transduction proteins inside the cell, leading to a specific cellular response) ;
7. (S) non-polar R groups of amino acid residues on the receptor form hydrophobic interactions with non-polar hydrocarbon tails of the membrane phospholipid molecules ;
AND
8. (F) G-protein linked receptors are embedded in (and span) the plasma membrane, (held by weak hydrophobic interactions) ;

A MP 1+4

The glucagon receptor is a Class B GPCR that plays a central role in the regulation of blood glucose levels and glucose homeostasis. The actions of glucagon are transduced via activation of the glucagon receptor.

(b) Describe how binding of glucagon aids in the maintenance of a constant blood glucose level. [4]

1. **binding of glucagon to specific binding site of GPCR induces a conformational change in the receptor ;**
2. **The receptor now binds to G protein and activates it ;**
3. **A molecule of GTP replaces the GDP on the G protein ;**
4. **The activated G-protein dissociates from the receptor and activates adenylyl cyclase ;**
5. **Adenylyl cyclase catalyses the conversion of ATP to / synthesis of cyclic AMP (cAMP) ;**
6. **The cAMP acts as a second messenger and triggers downstream signalling events such that glycogen phosphorylase is activated ;**
7. **Glycogen phosphorylase will catalyse the breakdown of glycogen to glucose (glycogenolysis) so that the blood glucose concentration increases and return back to constant/normal levels ;**

max 3 for MP 1-6

(c) The following findings were concluded from studies on the expression of the glucagon receptor mRNA in liver, adipose tissues and pancreatic islets.

- The promoter of the glucagon receptor gene contain regulatory elements for glucose and cyclic AMP.
- The expression of the glucagon receptor mRNA is stimulated by glucose and inhibited by cyclic AMP in liver.
- The number of glucagon receptors expressed in the liver can be regulated by glucagon.

Comment how the above features are advantageous to mammals. [1]

1. **allow/enable the expression of glucagon receptor gene in response to the presence/absence of glucose and cyclic AMP;**
2. **glucagon receptors are only expressed when required to increase efficiency / conserve/prevent the waste of energy and resources;**

[Total: 9]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 (a) The organelles of the endomembrane system in eukaryotic cells are related through direct contact or by the transfer of membrane segments as vesicles.

Outline the functions of the organelles of the endomembrane system and state the structural similarities between these organelles. [15]

(Max 12 from pt 1 – 17)

Functions of the rough ER

1. sites of protein synthesis ;
2. biochemical modification takes place in cisternal space / proteins may be modified by enzymes in the cisternal space of ER that add carbohydrate chains (or lipids) to them – glycosylation ;
3. serves as the intracellular transport system which transports the synthesised/modified proteins to other compartments within the cell by transport vesicles budding off from the ER membrane ;
4. The membrane of transport vesicles (formed from rough ER membrane) also replenishes the membrane of Golgi body ;

Functions of the smooth ER

5. responsible for the synthesis of lipids (e.g. phospholipids and steroid hormones) ;
6. involved in carbohydrate metabolism in the liver (glycogenolysis) ;
7. detoxification of drugs and other toxic substances in the liver ;
8. storage of calcium ions necessary for muscle contraction ;

Function of Golgi body

9. Further modify, sort and package proteins / products of ER into vesicles before transporting to other parts of the cell and/or out of the cell / targeted for other destinations ;
10. Formation of secretory vesicles containing matured proteins which move to the cell surface and fuse with the plasma membrane, releasing their contents out of the cell via the process of exocytosis ;
11. Formation of lysosomes containing hydrolytic enzymes/proteases/nucleases/lipases ;

(In plant cells)

12. Packaging of polysaccharides destined for secretion, e.g. mucus ;
13. Formation of secretory vesicles containing polysaccharide which move to the cell surface and fuse with the plasma membrane, releasing their contents out of the cell via the process of exocytosis ;
14. During the synthesis of new plant cell walls, transport vesicles containing cell wall materials/pectins/certain non-cellulose polysaccharides fuse to form the cell plate ;

MP 10&s13: award once

Function of Lysosomes

15. Intracellular digestion - to digest material which the cell consumes from the environment through phagocytosis ;
16. Autophagy - to digest parts of the cell such as damaged or worn-out organelles;
17. Autolysis - self-destruction of a cell by releasing the hydrolytic enzymes of all lysosomes within the cell ;

Structural similarities

18. Presence of (single) membrane / phospholipids bilayer ;
19. Presence of proteins/enzymes within organelles ;
20. Presence of fluid-filled space ;
21. Association with cytoskeleton ;

QWC:

Good spread of knowledge communicated without ambiguity to include:

At least 2 organelles' functions and at least 1 MP on the structural similarities.

- (b) Binary fission is a process where a single bacterial cell divides into two genetically identical daughter cells. Discuss if human cells are genetically identical and contrast the organisation of bacterial and human genome. [10]

Definition of genetically identical ;

1. daughter cells which contain exact copies/same genetic material or information/same number and type of chromosomes/same amount and type of DNA, as the parent cell

Human/eukaryotic cells can be considered genetically identical because:

2. (idea of) following fertilisation, single-cell zygote can undergo mitosis and divide into two genetically-identical cells ;
3. during S phase of interphase ; DNA replication takes place (with an identical copy of the DNA molecule synthesized) ;
4. most somatic cells of human are genetically identical ;

Not all cells of a human are genetically identical because:

5. Human is a multicellular organism with many different cell types ;
6. number of chromosomes are different in gametes/sperm/egg compared to most somatic cells ;
7. somatic hypermutation occurs in T and B cells ;
8. cells with specialised function such as red blood cells do not have nucleus ;
9. telomeres shortening takes place with repeated rounds of cell division, changing the DNA sequences ;
10. ref to virally infected cells where the viral DNA has been integrated into the chromosome of human host cell ;
11. ref to cancer cells with altered genes ;

Feature of comparison		Bacterial genome	Human genome
12.	No. of chromosomes/bases ;	Smaller number of chromosomes/bases	Greater number of chromosomes/bases
13.	No. of chromosomal set ;	Single chromosome/haploid/1 set	Multiple chromosomes/diploid
14.	Clustering of genes ;	Polycistronic genes / operons	Monocistronic genes
15.	Presence of distal control elements/enhancer/silencer ;	Absence of enhancer/silencer Presence of operator	Presence of enhancers / silencers
16.	Introns ;	Absent	Present
17.	Association with histones and other proteins ;	No association with histones/association with H-NS proteins	Association with histones / scaffolding proteins

QWC:

clearly expressed and well structured, using correct terminology to address both parts of the question ;

- 5 (a) Variation exists in individuals of the same species in a population due to a number of different reasons. Describe what causes variation and why it is important in natural selection. [15]

Causes of Variation (Max 12)

1. Variation is due to mutation, meiosis and sexual reproduction ;

Gene Mutation

2. Gene mutations are defined as changes in the sequence of DNA nucleotides in the gene;
3. Due to exposure to chemical carcinogens such as tobacco in cigarette smoke / ionising radiation such as UV-rays, X-ray / viruses / errors during DNA replication or repair etc.;
4. Base pair substitution / addition / deletion may occur, resulting in nonsense / silent / missense / frameshift mutation; (max 1)
5. Base-pair substitution – replacement of one nucleotide base pair with another base-pair in a gene, resulting in missense mutation/nonsense mutation/significant change in the encoded protein OR silent mutation/no/little effect on the encoded protein;
6. Base-pair addition – insertion of one or more nucleotide base pairs in a gene, resulting in frameshift mutation, whereby the reading frame is altered / nucleotides downstream from the mutation are improperly grouped into incorrect codons and read in different sets of threes;
OR
7. Base-pair deletion – loss of one or more nucleotide base pairs in a gene, resulting in frameshift mutation, whereby the reading frame is altered / nucleotides downstream from the mutation are improperly grouped into incorrect codons and read in different sets of threes;

Max 2 for MP 4-7

Chromosomal aberration

8. Chromosomal aberration can be categorised as numerical aberration (the change in chromosome number) or structural aberration to chromosomes;
9. Translocation – a section of chromosome breaks off from one chromosome and becomes attached to another chromosome;
10. Duplication – a section of a chromosome replicates such that a set of gene loci is repeated;
11. Deletion – a chromosome breaks at two points, the middle portion is displaced with the two ends joining together;
12. Inversion – a chromosome breaks at two locations and the middle portion flips through 180° before rejoining;
13. Aneuploidy is a condition in the nucleus where there are one or several chromosomes more than or less than the diploid number of chromosomes;
14. Aneuploidy can result from non-disjunction during anaphase / when a haploid gamete fuses with a gamete carrying n-2, n-1, n+1 or n+2 chromosomes;
15. Polyploidy is a condition of the nucleus where there are three or more times the haploid number of chromosomes, e.g. 3n, 4n and 5n;
16. It can result from non-disjunction, the fusion of a diploid gamete with a normal haploid gamete giving a triploid nucleus;

Meiosis

- 17. In meiosis, crossing over between non-sister chromatids of homologous chromosomes occurs during prophase I;
- 18. Independent assortment of homologous chromosomes during metaphase I and chromatids during metaphase II;
- 19. Results in new combination of alleles;

Sexual reproduction

- 20. Random fusion of gametes during sexual reproduction;
- 21. Results in new combination of alleles (award once);

Genetic variation in prokaryotes (max 2 per mechanism)

- 22. Transformation is the uptake of naked, foreign DNA by competent cells from the surrounding environment resulting in the alteration of a bacterial cell's genotype and phenotype;
- 23. A (competent) bacterium takes up foreign DNA;
- 24. The foreign DNA is incorporated into bacterium's own DNA via homologous recombination/through crossing over with a homologous region found on the bacterial chromosome;
- 25. Transduction is the process by which bacterial DNA/genes is transferred from one bacterium (host cell) to another (recipient cell) via a bacteriophage;
- 26. When a (virulent) bacteriophage undergoes lytic cycle, a small piece of the host bacterial cell's degraded DNA is packaged within the capsid (of a defective phage); OR
- 27. When a (temperate) bacteriophage enters into lytic cycle from lysogenic cycle, a small region of the host bacterial DNA that was adjacent to the prophage is excised and the phage-host hybrid DNA is packaged within the capsid (of a defective phage);
- 28. The defective phage infects another bacterial cell and inject the piece of host bacterial DNA/ phage-host hybrid DNA into the newly infected bacterial cell cytoplasm, the host bacterial DNA/ phage-host hybrid DNA is incorporated into bacterium's DNA via homologous recombination/through crossing over;
- 29. Conjugation is the direct transfer of genetic material from one bacterial cell to another, through a temporary link between two cells (requires cell to cell contact);
- 30. F⁺ cell/donor bacterial cell with F factor produces sex pilus to attach itself to F⁻ cell/recipient cell;
- 31. A temporary cytoplasmic mating bridge is formed between the two bacterial cells which allows F⁺ cell to transfer its F plasmid to the F⁻ cell (by rolling circle mechanism);

Importance in Natural Selection

- 32. Variation describes the differences in characteristics / means the presence of different characteristics;
- 33. due to presence of different alleles in the different individuals;
- 34. resulting in differential reproductive success / different survival rates;
- 35. Variations in characteristics are subjected to selection pressure from the environment;
- 36. There can be continuous / discontinuous variation;
- 37. Due to interaction of genotype and environment;
- 38. variants with favourable characteristics will survive to maturity, reproduce and pass down their favourable alleles to their offspring;
- 39. Those with unfavourable characteristics die and fail to do so;

QWC:

Good spread of knowledge communicated without ambiguity to include:

At least 2 causes of variation and at least 1 MP on the importance.

- (b) Discuss, using known examples, how limiting factors can influence the rate of various biological processes. [10]

Definition

1. A limiting factor is that factor which directly affects/determines the rate of reaction/process if its quantity is changed / when the factor is in lowest/shortest/scarcest supply. ;
[Reject “in short / shorter / low / lower supply”; MUST be “shortest” or “lowest”]

Biological Processes

2. Biological Process: Enzyme-catalysed Reactions ;

Factor	How Factor Influences Rate	
3. Substrate concentration	+ illustrate how factor influence rate	;
4. Enzyme concentration		;
5. Temperature		;
6. pH		;

REJECT: Concentration of competitive or non-competitive inhibitors as limiting factors.

7. Biological Process: Facilitated Diffusion ;

Factor	How Factor Influences Rate	
8. (Steepness of) concentration gradient / difference in concentrations of transported substance across a cell / biological membrane	+ illustrate how factor influence rate	;
9. Density of channel/carrier proteins on the membrane		;

10. Biological Process: Photosynthesis ;

Factor	How Factor Influences Rate	
11. Light intensity	+ illustrate how factor influence rate	;
12. Temperature		;
13. Carbon dioxide concentration		;

14. Biological Process: Aerobic Respiration ;

Factor	How Factor Influences Rate	
15. Glucose / hexose sugars concentration	+ illustrate how factor influence rate	;
16. Oxygen availability		;
17. Temperature		;
18. NAD ⁺ / FAD ⁺ / mobile electron carriers availability		;

19. Biological Process: Population Growth / Bacteria Colony Growth ;

Factor	How Factor Influences Rate	
20. Nutrient availability	+ illustrate how factor influence rate	;
21. Water availability		;
22. Temperature		;
23. Competition for resources / Predation / AVP		;

REJECT: Signal Reception / Ligand-Receptor Interactions as a form of biological process that is influenced by limiting factors. Any limits on rates of signal propagation and hence response efficacy is overcome by signal amplification.

QWC:

Clearly communicates limiting factors of 2 different biological processes